

UIT2734
IOT ARCHITECTURES AND
APPLICATIONS

Unit II: IoT Protocols

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Communication Criteria

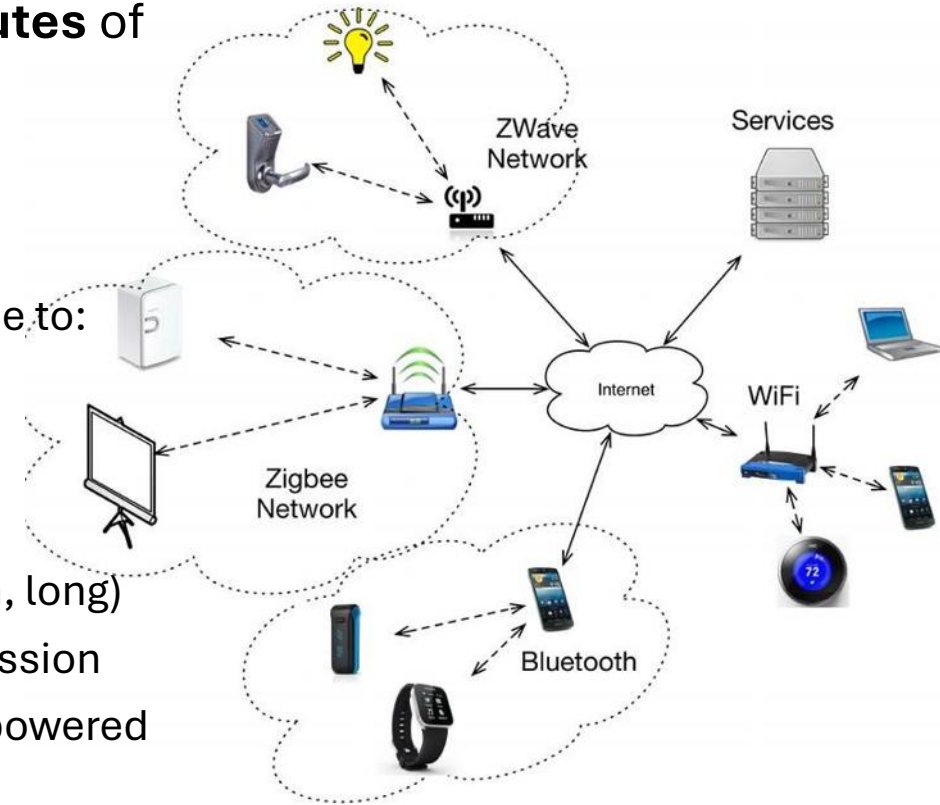
➤ Communication criteria describe the **characteristics and attributes** of IoT access technologies

➤ **Wired vs. Wireless:**

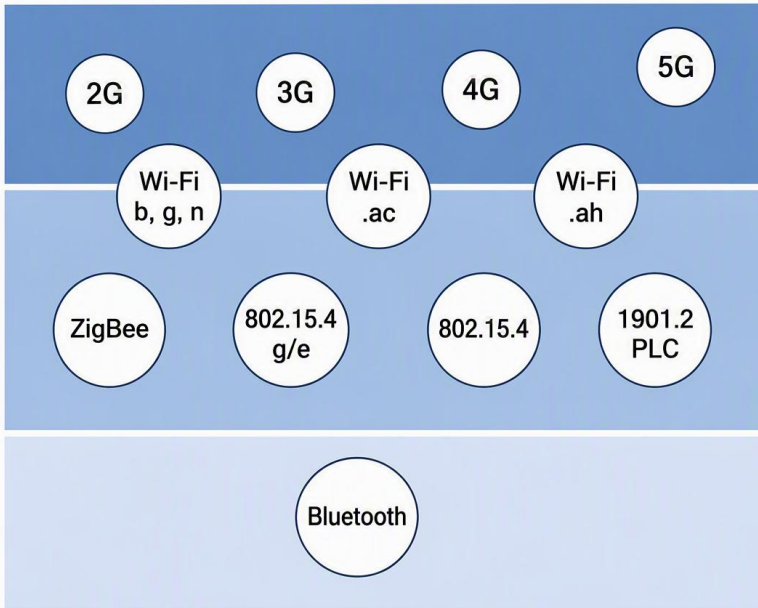
- IoT access technologies could be **wired** or **wireless**
- **Wireless communication is prevalent** for smart object connectivity due to:
 - Ease of deployment
 - Allows smart objects to be **mobile** – moving without losing connectivity

➤ **Few Basic Communication Criteria:**

- **Range** – Distance over which devices can communicate (short, medium, long)
- **Frequency Bands** – Licensed vs. unlicensed spectrum used for transmission
- **Power Consumption** – Energy requirements (grid-powered vs. battery-powered nodes)
- **Topology** – Network structure (star, mesh, peer-to-peer, cluster tree)
- **Constrained Devices** – Limited resources (RAM, flash, processing power) of IoT nodes
- **Constrained-Node Networks** – Low-power and lossy networks (LLNs) with limited throughput



Range



- Range refers to the **maximum distance** over which two devices can successfully communicate.
- **Short Range** – Tens of meters between two devices
 - Often considered as an alternative to serial cable
 - Examples: Bluetooth (IEEE 802.15.1), Visible Light Communications (IEEE 802.15.7)

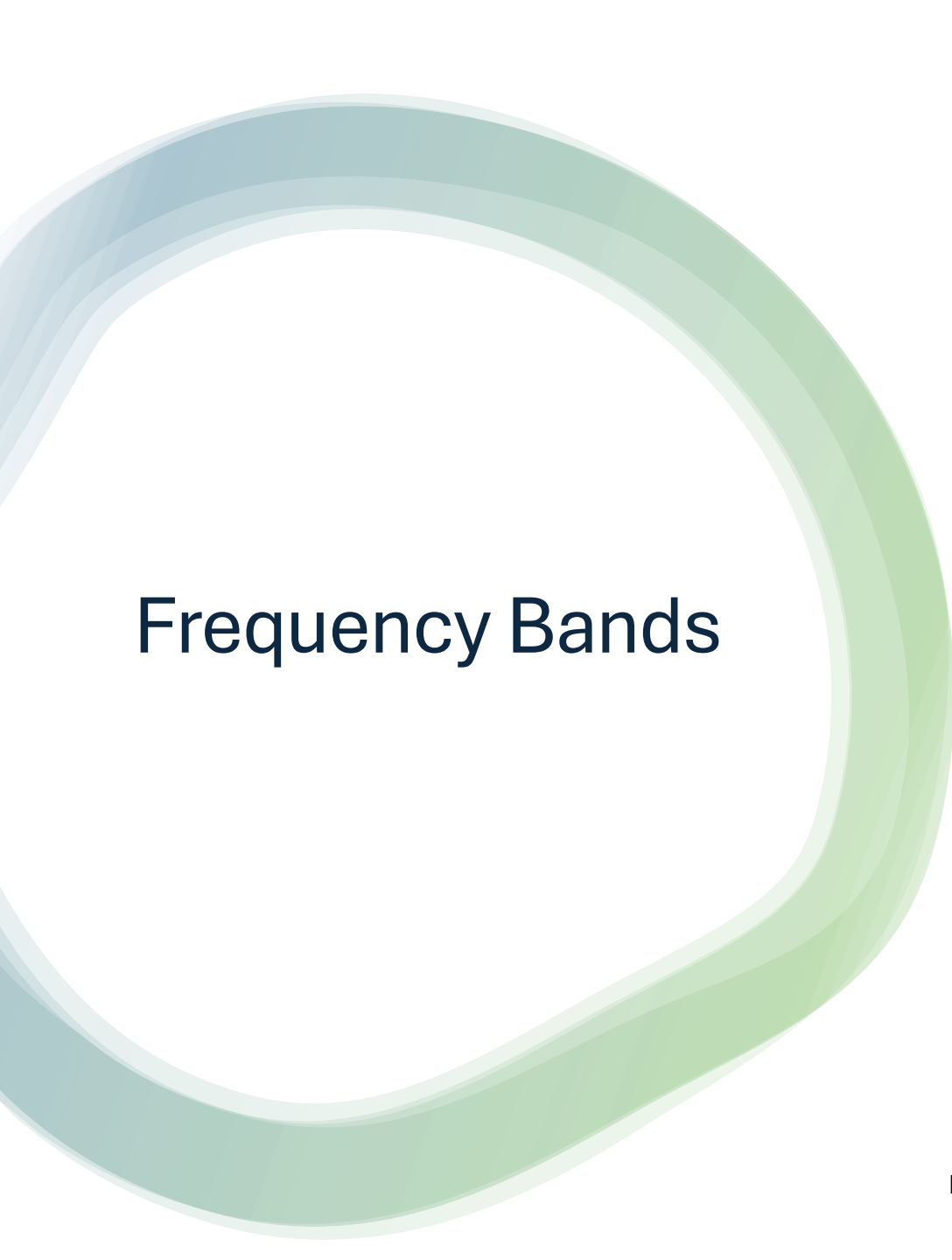
- **Medium Range** – Tens to hundreds of meters between two devices

- Long Range** ➤ Wireless examples: Wi-Fi (IEEE 802.11), IEEE 802.15.4 (LR-WPAN), IEEE 802.15.4g (SUN)
 - Wired examples: Ethernet (IEEE 802.3), Narrowband PLC (IEEE 1901.2)
- Medium Range**

- **Long Range** – Greater than 1 mile (1.6 km) between two devices

- Short Range** ➤ Wireless examples: 2G/3G/4G Cellular, IEEE 802.11ah, Low-Power Wide-Area (LPWA)
 - Wired examples: Ethernet over optical fiber, Broadband PLC

- The choice of access technology is primarily driven by required transmission range – different use cases demand different ranges



Frequency Bands

- Frequency bands refer to the **specific ranges of radio spectrum** used for wireless communication between IoT devices
- **Radio Spectrum Regulation:**
- Radio spectrum is **regulated by countries and/or organizations**
- Examples of regulatory bodies:
 - **ITU** – International Telecommunication Union (global)
 - **FCC** – Federal Communications Commission (USA)
 - **TRAI** – Telecom Regulatory Authority of India (India)

Frequency Bands

- Frequency Bands are Split Between:

- **1. Licensed Spectrum**

- Applicable to **long-range access technologies**

- Users must **subscribe to services**

- Common licensed spectrum for IoT:

- Cellular (900–2100 MHz)

- NB-IoT (700–900 MHz)

- **2. Unlicensed Spectrum**

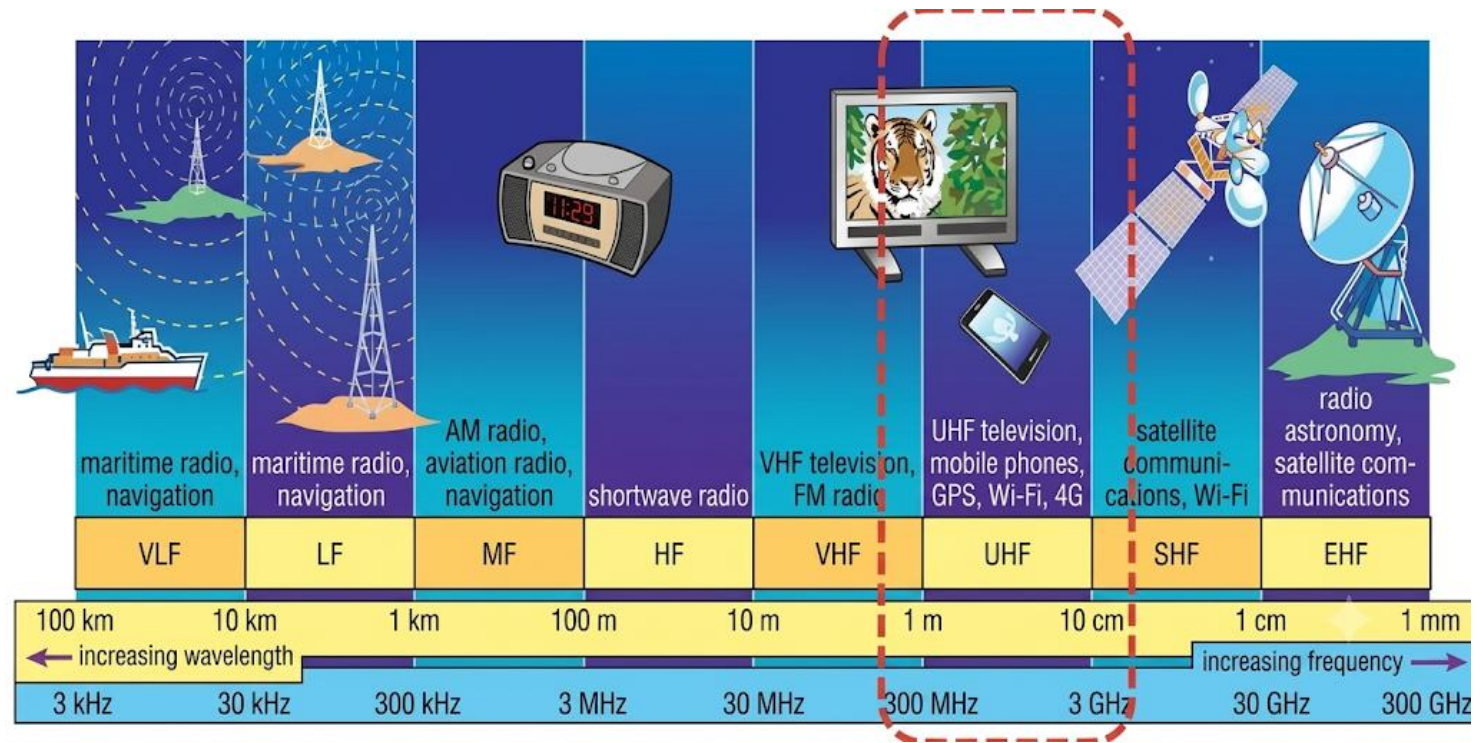
- **No guarantees or interference protections** are offered

- Uses **ISM** (Industrial, Scientific, and Medical) portions of radio bands

- Well-known ISM bands for IoT:

- **2.4 GHz, 5 GHz, 915 MHz** – used by Wi-Fi, BLE, ZigBee

- **868 MHz** – used by LoRa



Licensed spectrum offers reliability with a cost; unlicensed spectrum is free to use but prone to interference.

Power Consumption

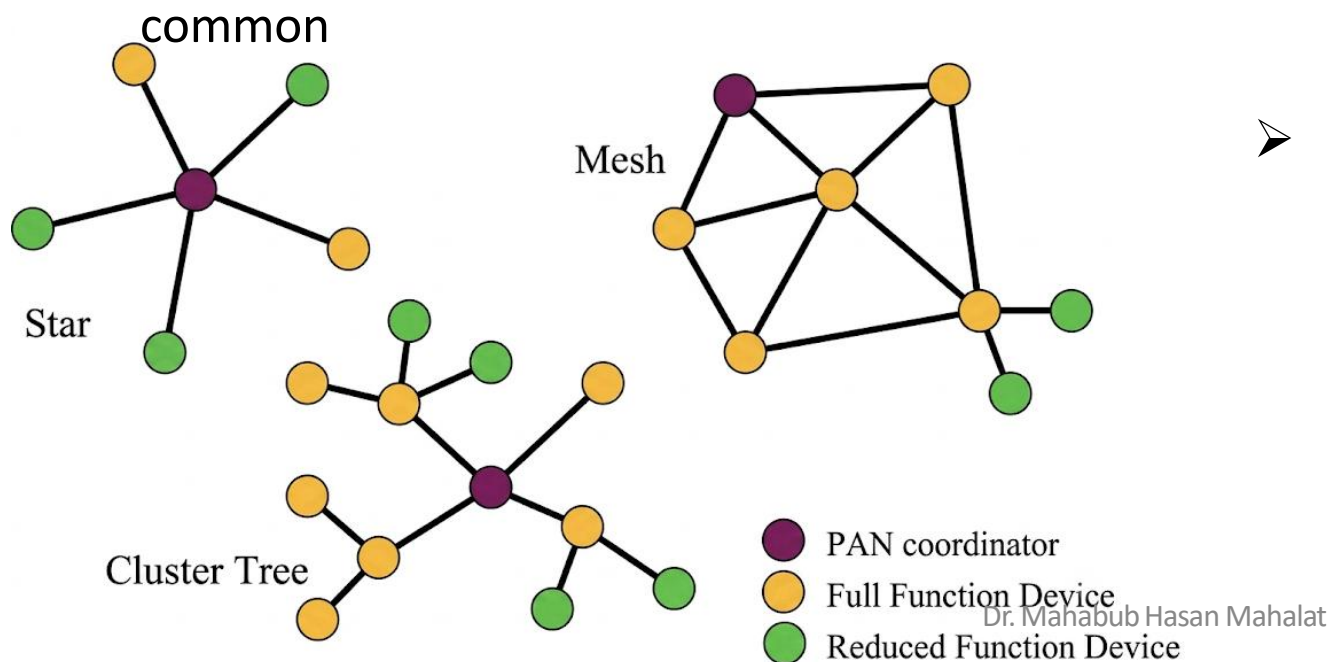
- Power consumption refers to the energy requirements of IoT devices – a critical factor that determines battery life and deployment feasibility.
- **Types of Power Sources:**
- **Grid-Powered Node**
 - Node has a direct connection to a grid power source
 - Communications are **not limited** by power consumption criteria
 - Ease of deployment is **limited by availability** of a power source
 - Makes mobility **more complex**
- **Battery-Powered Nodes**
 - Bring **more flexibility** to IoT devices
 - Batteries are **small in size**
 - Batteries can be **changed or recharged**
- **Key Challenge:**
- Most IoT nodes are battery-powered
- Wireless access technologies must address:
 - The needs of low power consumption
 - Connectivity for battery-powered nodes

Power Consumption Comparison of IoT Access Technologies

Technology	Bluetooth	ZigBee	WiFi	LoRaWAN	NB-IoT
Standard	IEEE 802.15.1	IEEE 802.15.4	IEEE 802.11b	LoRaWAN	3GPP NB-IoT
Sleeping	9 μ A	12 μ A	30 μ A	0.1 μA	3 μ A
Awake/Idle	35 mA	50 mA	245 mA	1.4 mA	6 mA
Transmitting	39 mA	52 mA	251 mA	44 mA	220 mA
Receiving	37 mA	54 mA	248 mA	12 mA	46 mA
Power Supply	3.3 V	3.3 V	5 V*	3.3 V	3.6 V

Topology

- Three main topology schemes are dominant:
 - star, mesh, and peer-to-peer (cluster tree)
- For long-range and short-range technologies:
 - star topology is prevalent
- For medium-range technologies:
 - star, peer-to-peer, or mesh topology is



- IEEE 802.15.4, 802.15.4g, and wired IEEE 1901.2a PLC are generally deployed as a **mesh topology**.
- Indoor Wi-Fi deployments are mostly **star topologies**
- Full Function Device (FFD) :
 - A node that implements the **full network functions**
 - Can communicate with **FFD and RFD** devices.
 - FFD Example: Zigbee Coordinator, Router
- Reduced Function Device (RFD):
 - The device can implement a **subset of protocol functions** to perform just a specialized part (communication with the coordinator).
 - RFD Example: Temperature sensor, Motion sensor, Door sensor

Constrained Devices

- Constrained nodes have limited resources that impact their networking feature set and capabilities.
- RFC 7228 defines **three classes for constrained nodes**: Class 0, 1, 2

	RAM	Flash Storage	IP stack	Security Scheme	Example
Class 0	< 10 KB	< 100 KB	Not present	No	Push button
Class 1	> 10 KB	> 100 KB	Optimized IP stack	Light	Sensors
Class 2	> 50 KB	> 250 KB	Full IP stack	Yes	Smart meter

Constrained Networks

- Constrained-node networks are often referred to as low-power and lossy networks (LLNs)
- **Layer 1 and Layer 2** protocols must be evaluated using the following characteristics:
 - Data rate and throughput
 - Latency and determinism
 - Overhead and payload.
- **Data rate & throughput:**
 - Data rates available from 100 bps to tens of Mbps
 - Actual throughput is less, sometimes much less, than the data rate
- **Latency & determinism:**
 - When latency is a strong concern, emergent access technologies such as Time-Slotted Channel Hopping (TSCH) mode of IEEE 802.15.4e should be considered.
- **Overhead & Payload**
 - The minimum IPv6 MTU size is expected to be **1280 bytes**.
 - MTU size for IEEE 802.15.4 is **127 bytes**; payload in LoRaWAN may **be from 19 to 250 bytes**
 - So, the fragmentation of the IPv6 payload has to be performed by the link layer

IoT Access Technologies



These are the **physical and data link layer** technologies that connect smart objects (sensors, actuators) to the network.



They define **how** devices communicate, **how fast**, **how far**, and **how securely**



Key Layers in Focus:

Layer	Function	Analogy
Physical Layer (PHY)	Defines the physical medium (wireless/wired), frequencies , data rates , and modulation techniques	The road and traffic rules for data
MAC Layer	Manages access to the physical medium , collision avoidance , and addressing	The traffic controller coordinating who transmits

Why So Many Technologies?

Different IoT use cases have different requirements:

Range – Few meters vs. several kilometers

Data Rate – Few bytes/hour vs. streaming video

Power – Battery-powered (years) vs. mains-powered

Cost – Ultra-low-cost vs. higher-cost

Environment – Indoor, outdoor, industrial, smart city, etc

Technologies Covered:

Technology	Category	Best For
IEEE 802.15.4	WPAN (Wireless Personal Area Network)	Foundational, low-power, short-range WSNs
IEEE 802.15.4g/e	WPAN (Smart Utility Networks)	Smart grid, utilities, industrial IoT
IEEE 1901.2a	Power Line Communication (PLC)	Smart grid over existing power lines
IEEE 802.11ah (Wi-Fi HaLow)	WLAN (Sub-1 GHz Wi-Fi)	Long-range Wi-Fi for IoT
LoRaWAN	LPWAN (Low-Power Wide-Area)	Long-range, low-power, battery-operated sensors



IEEE 802.15.4 – Introduction & Standardization

- IEEE 802.15.4 is a wireless access technology for **low-cost and low-data-rate devices** that are powered or run on batteries.
- It enables easy installation using a **compact protocol stack** while remaining simple and flexible.
- Several network communication stacks leverage this technology to address a wide
- Range of IoT use cases
- **Common Deployments:**
 - Home and building automation
 - Automotive networks
 - Industrial wireless sensor networks
 - Interactive toys and remote controls

Standardization and Alliances of IEEE 802.15.4

Aspect	Details
Standards Body	IEEE 802.15 Task Group 4 – defines low-data-rate PHY and MAC layer specifications for Wireless Personal Area Networks (WPAN)
First Publication	2003; multiple iterations published since then
Evolution	Newer releases supersede older ones, integrate addendums, and add features/clarifications
Upper Layers	The PHY and MAC layers are foundations for several networking protocol stacks – upper layers are different and promoted separately through various organizations

Protocol Stacks Based on IEEE 802.15.4

- **ZigBee's Role:**
- ZigBee utilizes the IEEE 802.15.4 Standard at the lower PHY and MAC layers, and specifies the network, security, and application support layers on top.
- The ZigBee Alliance (now Connectivity Standards Alliance) maintains the ZigBee specification

Protocol	Description
ZigBee	Promoted through the ZigBee Alliance; defines upper-layer components (network through application) and application profiles (building automation, home automation, healthcare)
6LoWPAN	IETF standard describing how to transport IPv6 packets over IEEE 802.15.4 layers
Thread	Another protocol stack built on 802.15.4

IEEE 802.15.4 – Physical Layer

- Defines the **frequency bands, modulation, and radio parameters**
- Handles **data transmission** via a physical radio transceiver
- Frequency Bands and Data Rates:

Frequency Band	Channels	Data Rate	Regions
868 MHz	1 channel	20 kbps	Europe
915 MHz	10 channels	40 kbps	Americas
2.4 GHz	16 channels	250 kbps	Worldwide (ISM band)

- **Modulation:**
 - **Direct-Sequence Spread Spectrum (DSSS)** with **Offset QPSK** at 2.4 GHz
 - Low-power spread spectrum radio operation

IEEE 802.15.4 – MAC Layer

➤ **MAC Layer (Medium Access Control) – Functions:**

- Bridges the **physical layer** with **data link control**
- Manages **channel access** and **frame formats**
- Defines **addressing** and **frame types**

• **Channel Access Methods:**

Mode	Description
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance – device "listens" before transmitting; if channel is busy, a random wait time occurs
Beacon-Enabled Mode	Time is subdivided into superframes ; uses slotted CSMA/CA
Non-Beacon-Enabled Mode	Uses unslotted CSMA/CA

IEEE 802.15.4 – MAC Layer...



Superframe Structure (Beacon Mode):

Bounded by periodic **beacon frames** from the coordinator

Divided into **active** and **inactive** periods

Active period uses CSMA/CA procedure for communication



Frame Types:

Beacon frame

Data frame

Acknowledgment frame

MAC command frame

IEEE 802.15.4 – Topology & Security

- Supported Topologies :

Topology	Description	Characteristics
Star	All devices communicate through a central coordinator	Simple, single-hop; PAN coordinator forms links with multiple devices
Peer-to-Peer	Devices communicate directly with each other	Enables more complex topologies
Mesh	Routers relay messages through multiple hops; packets can take multiple paths	Extends range; routes around failed nodes
Cluster-Tree	Hierarchical combination of star networks	Covers more area; fixes maximum routers and tree depth

IEEE 802.15.4 – Topology & Security...

➤ Security:

Aspect	Details
Cipher	Advanced Encryption Standard (AES) block cipher
Key Size	128, 192, or 256 bits
Security Modes	Based on AES block cipher: • Null – neither encryption nor authentication (default) • Authentication-only – AES-CBC-MAC • Encryption + Authentication – AES-CCM
Authentication Tag	32, 64, or 128 bits
Access Control	Access Control List (ACL) to control which security suite and keys to use
ZigBee Security	AES-128 encryption at network and application layers

IEEE 802.15.4 – Limitations

Later variants of 802.15.4 (g/e) address these issues

Limitation	Description
MAC Reliability	CSMA/CA can cause unbounded latency
Unbounded Latency	No guarantee on maximum delay
Susceptibility to Interference	Lacks frequency-hopping technique
Multipath Fading	Signal reflections cause degradation

IEEE 802.15.4 Competitive Technologies:

Technology

Comparison with IEEE 802.15.4 / ZigBee

Bluetooth Low Energy (BLE)

Similar low power consumption; shorter range; different application focus

6LoWPAN

IPv6 adaptation layer over 802.15.4; enables IP connectivity

Thread

Low-power mesh implementation; based on 802.15.4; focuses on reliability and scalability

Wi-Fi (802.11)

Higher data throughput; higher power consumption; not suitable for battery-powered sensors

Z-Wave

Similar to ZigBee; used in smart home; different frequency bands

Market Trends:

- Shift from ZigBee to **Wi-Fi 6** for improved data throughput and lower latency
- Shift from Bluetooth Classic to **Bluetooth Low Energy (BLE)** for enhanced power efficiency
- Shift from Z-Wave to **Thread** for better scalability and interoperability in smart home applications

ZigBee

- ZigBee is a protocol stack built on the IEEE 802.15.4 standard
- Designed for low-bandwidth and low-power smart objects
- Maintained by the **ZigBee Alliance** (now Connectivity Standards Alliance)
- **History and Evolution:**
 - First ZigBee specification ratified in 2004 (shortly after IEEE 802.15.4)
 - Industry support: 100+ companies at launch → 400+ member companies today
- **Key revisions:**
 - 2006 Revision – Introduced "clusters" (sets of commands for specific functions)
 - 2007 (ZigBee Pro) – Expanded clusters for metering, temperature, lighting control

ZigBee...

➤ What are Clusters?

- **Clusters** = Sets of commands and message types for specific functions
- Clusters from different functional domains form **application profiles**
- Vendors implementing predefined profiles ensure interoperability

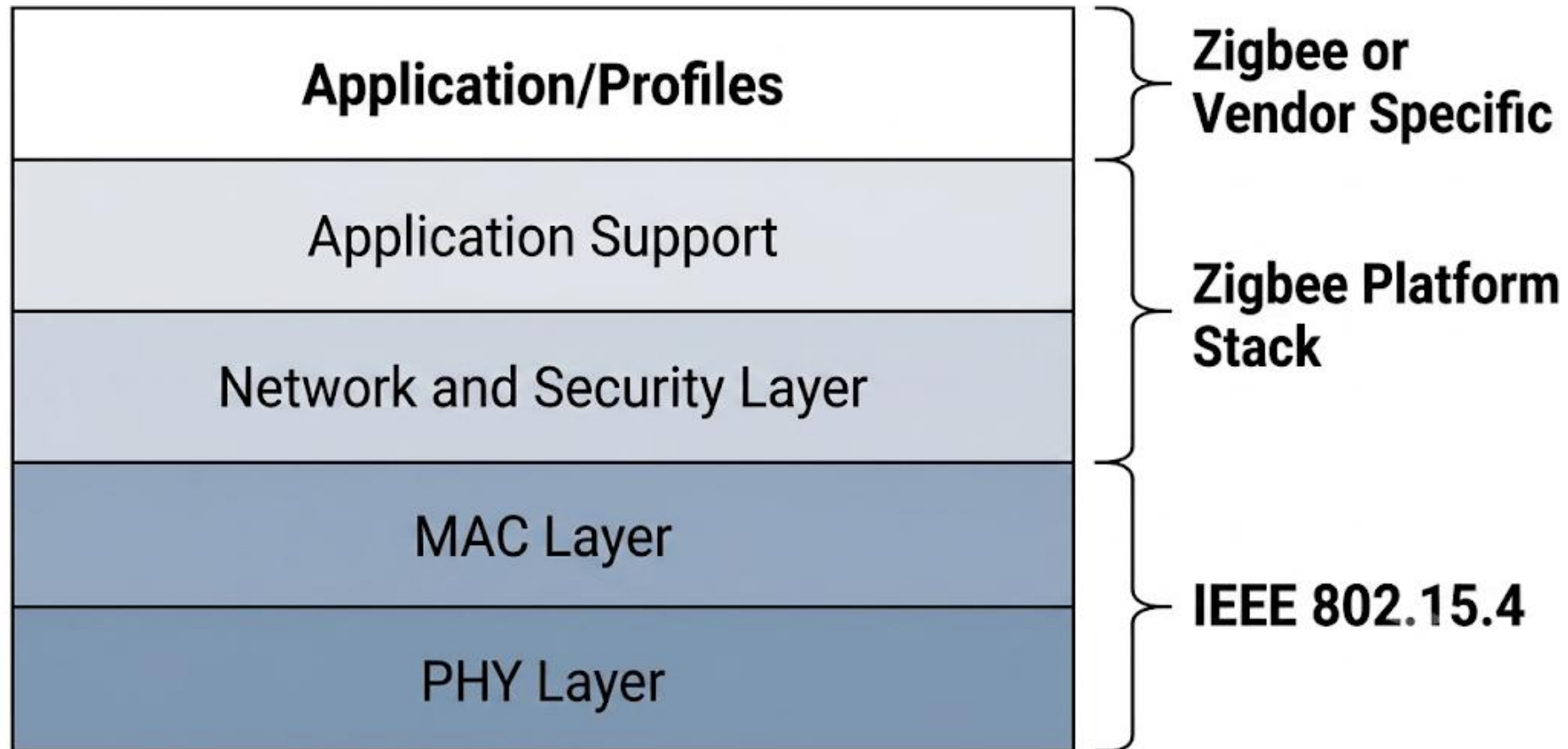
➤ Popular Application Profiles:

- **Home Automation** – Lighting, thermostats, security
- **Smart Energy** – Smart meters, utility monitoring/control

• ZigBee Main Application Areas:

Domain	Applications
Commercial/Retail Automation	Temperature, humidity, asset tracking
Home Automation	Lighting, thermostats, security
Smart Energy	Smart meters, utility delivery control

ZigBee Protocol Stack



ZigBee Protocol Stack...

- **ZigBee Stack Architecture (Traditional):**

Layer	Function
Application Layer	Application profiles (Home Automation, Smart Energy, custom)
Application Support Layer	Interfaces network layer with applications
Network & Security Layer	Network startup, configuration, routing, securing communications
MAC Layer (IEEE 802.15.4)	Channel access, frame types, addressing
PHY Layer (IEEE 802.15.4)	Frequency bands, modulation, data rates

Key Features of ZigBee

Feature	Description
Lower Layers	Uses IEEE 802.15.4 PHY and MAC
Network Layer	Supports star, tree, and mesh topologies
Security	AES-128 encryption at multiple layers
Application Profiles	Predefined (Home Automation, Smart Energy) and custom
Interoperability	ZigBee Alliance certification ensures vendor interoperability

ZigBee IP – The Evolution

- **ZigBee IP** embraces open IETF standards for **low-power, low-bandwidth IoT communications** while maintaining IEEE 802.15.4 at the lower layers.
- **ZigBee IP Stack Architecture:**

Layer	Protocol
Application	ZigBee-Specific Layers (SE 2.0, custom apps)
Transport	TCP (connection-oriented) and UDP (connectionless)
Network	IPv6, ICMPv6, RPL (routing), 6LoWPAN ND
Adaptation	6LoWPAN (fragmentation, header compression)
MAC Layer	IEEE 802.15.4
PHY Layer	IEEE 802.15.4

ZigBee IP (Smart Energy 2.0 Profile)	
UDP	TCP
IPv6, ICMPv6, 6LoWPAN-ND	RPL
6LoWPAN Adaptation Layer	
802.15.4-2006 MAC	
802.15.4-2006 PHY	

Key Features of ZigBee IP

Feature	Description
IPv6 Support	All nodes support IPv6, ICMPv6, and 6LoWPAN ND
6LoWPAN	Provides fragmentation and header compression; supports mesh-over or route-over forwarding
RPL (Routing Protocol for LLNs)	Used for routing across mesh networks
TCP/UDP Support	Both connection-oriented and connectionless services
IETF Standards	Built on open standards at every layer under the application layer

➤ Why ZigBee IP is Important:

- Designed specifically for **Smart Energy (SE) Profile 2.0** (smart metering, residential energy management)
- Can be used for any application needing a **standards-based IoT stack**
- Enables **interoperability** with other IoT solutions built on open standards
- No longer locked into proprietary ZigBee-only networks – can integrate with any 802.15.4 network using open IoT standards

Comparison – Traditional ZigBee vs. ZigBee IP

Aspect	Traditional ZigBee	ZigBee IP
PHY/MAC	IEEE 802.15.4	IEEE 802.15.4
Network Layer	ZigBee-specific	IPv6 + RPL (IETF standards)
Adaptation Layer	Not present	6LoWPAN (fragmentation, header compression)
Transport Layer	ZigBee-specific	TCP/UDP
Application	ZigBee profiles (HA, SE 1.0)	Open + ZigBee profiles (SE 2.0)
Interoperability	Only within ZigBee ecosystem	Open – integrates with other IoT solutions
Key Use	Home automation, building automation	Smart Energy Profile 2.0, standards-based IoT

ZigBee IP represents a shift from a proprietary ZigBee stack to an open standards-based stack built on IETF protocols (IPv6, 6LoWPAN, RPL, TCP/UDP), enabling true interoperability with other IoT solutions.

IEEE 802.15.4g and 802.15.4e – Introduction

- IEEE 802.15.4g-2012 and IEEE 802.15.4e-2012 are **amendments** to the IEEE 802.15.4-2011 standard. They have been fully integrated into the core IEEE 802.15.4-2015 specification
- **Why Were These Amendments Needed?**
- The original IEEE 802.15.4 had limitations:

Limitation	Issue
Small Payload	Maximum (PHY Service Data Unit)PSDU size of 127 bytes – too small for IPv6 (1280 bytes MTU)
No Frequency Hopping	Susceptible to interference and multipath fading
Unbounded Latency	CSMA/CA could cause unpredictable delays
Limited Range	Not optimized for large outdoor deployments

802.15.4g – addresses **PHY layer** limitations for large outdoor networks

802.15.4e – addresses **MAC layer** limitations for industrial and critical applications

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IEEE 802.15.4g – Smart Utility Networks (SUN)

- IEEE 802.15.4g is a PHY amendment that optimizes **large outdoor wireless mesh networks** for **Field Area Networks (FANs)**. It is also known as the **Smart Utility Networks (SUN)** standard
- **Key PHY Enhancements**

Feature	802.15.4 (Original)	802.15.4g (SUN PHY)
Max PSDU/Payload	127 bytes	2047 bytes
CRC Error Protection	16-bit	32-bit
Frequency Bands	868 MHz, 915 MHz, 2.4 GHz	169 MHz to 2.4 GHz
PHY Mechanisms	Single (O-QPSK)	Multiple (FSK, OFDM, O-QPSK)

Key PHY Enhancements Over 802.15.4

Requirement	Conventional 802.15.4	Why 802.15.4g helps
Network size	Small/medium sensor networks	Designed for large outdoor FANs
Communication range	Typically short-range	Better suited to longer-range outdoor links
Packet size	Up to 127 bytes	Up to 2047 bytes
Frequency flexibility	Limited PHY choices	Sub-GHz + 2.4 GHz options
PHY options	Fewer choices	FSK, OFDM, O-QPSK
Error detection	16-bit FCS	Supports 32-bit FCS
Typical applications	Sensors, home/industrial IoT	Smart meters, grid monitoring, utility automation

➤ Why Larger Payload Matters:

- Default IPv6 MTU is **1280 bytes**
- With 802.15.4g, **fragmentation is no longer necessary** at Layer 2
- Reduces overhead and improves efficiency

Three PHY Mechanisms (Must Support at Least One)

- **802.15.4g provides three different ways to transmit data.**
A device needs to support **at least one** of these PHY options

PHY	Simple idea	Suitable when
MR-FSK	Sends data using different frequencies	Long range + low power are important
MR-OFDM	Sends data using many subcarriers simultaneously	Higher data rate / throughput is required
MR-O-QPSK	Similar to the PHY used in traditional 802.15.4	Low-cost compatibility with existing 802.15.4 technology is useful

- **Why three PHYs?**
 - Different utility applications have different requirements:
 - **Long distance + low power: MR-FSK**
 - **Higher data rate: MR-OFDM**
 - **Reuse existing 802.15.4 technology: MR-O-QPSK**

IoT Use Cases for 802.15.4g

- Designed for large numbers of distributed devices communicating over outdoor areas:
- **1. Smart Electricity Metering**
 - Thousands of **smart meters** periodically send readings
 - Data: energy consumption, voltage, current, meter status
 - Enables **remote meter reading and fault/tamper detection**
- **2. Smart Grid Monitoring**
 - Sensors monitor **transformers, substations and power lines**
 - Data: voltage, current, temperature and equipment status
 - Enables **remote monitoring and distribution automation**
- **3. Smart Street Lighting**
 - Each street light is connected to the wireless network
 - Monitor **ON/OFF status, energy consumption and faults**
 - Enables **remote control and automatic scheduling**

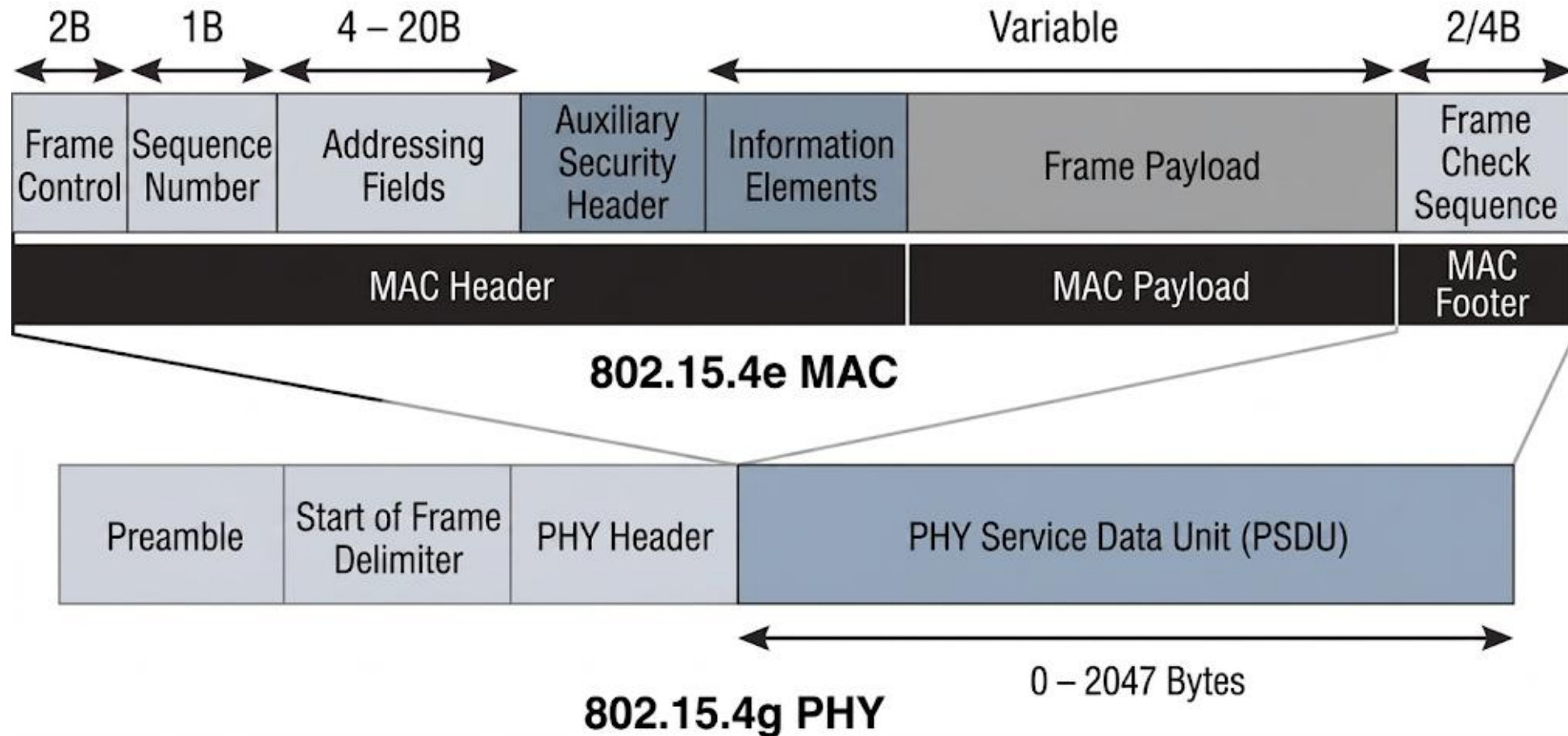
IEEE 802.15.4e – MAC Layer Enhancements

- IEEE 802.15.4e is a **MAC** amendment
- Adds new **MAC operation modes** to address the limitations of the original 802.15.4 MAC – particularly for **industrial and mission-critical applications**
- **Key MAC Enhancements**

Enhancement	Description
Time-Slotted Channel Hopping (TSCH)	Guarantees media access and channel diversity through synchronized time slots and frequency hopping
Information Elements (IEs)	Allows exchange of additional metadata at MAC layer using TLV format
Enhanced Beacons (EBs)	Extends beacon flexibility with application-specific content using IEs
Enhanced Beacon Requests (EBRs)	Allows selective request of information using IEs
Enhanced Acknowledgement	Integrates frame counter to protect against spoofing attacks

802.15.4e MAC enhancements are often paired with 802.15.4g PHY – together they form the foundation for Wi-SUN and industrial IoT deployments.

IEEE 802.15.4g/e MAC Frame Format



IEEE 802.15.4e MAC Enhancements in Details

➤ Time-Slotted Channel Hopping (TSCH):

Feature	Description
Concept	MAC operation mode that guarantees media access and channel diversity
Time Division	Time divided into fixed time slots – one packet + ACK per slot
Slot Frame	Group of time slots repeated regularly – provides guaranteed access
Channel Hopping	Uses different channels for transmission at different times
Synchronization	Transmitter and receiver agree on channels and timing via global time slot counter + global channel hopping sequence list
Benefits	• Guaranteed bandwidth • Predictable latency • Robustness in noisy environments • Smoother coexistence with other wireless technologies

IEEE 802.15.4e MAC Enhancements in Details

- **Information Element (IE)**
- **Optional IE field added to an IEEE 802.15.4 MAC frame.**
- It carries **additional information** needed by the MAC layer or other protocols.
- Think of an IE as an “**extra information box**” **attached to a frame.**
- **IEs** allow a frame to carry additional information without changing the basic MAC frame structure
- Example:
 - Normal data: Temperature = 28°C
 - Additional IE: Channel = 15 , Timing = ..., Network information = ...
- Two Main Types

Type	Location	Purpose
Header IE	MAC Header	Additional MAC-related information
Payload IE	MAC Payload	Additional information for MAC/protocol functions

IEEE 802.15.4e MAC Enhancements in Details

➤ **Enhanced Beacon (EB)**

➤ An **EB** is a more flexible beacon that can carry **additional information using Information Elements (IEs)**.

➤ **What can an EB carry?**

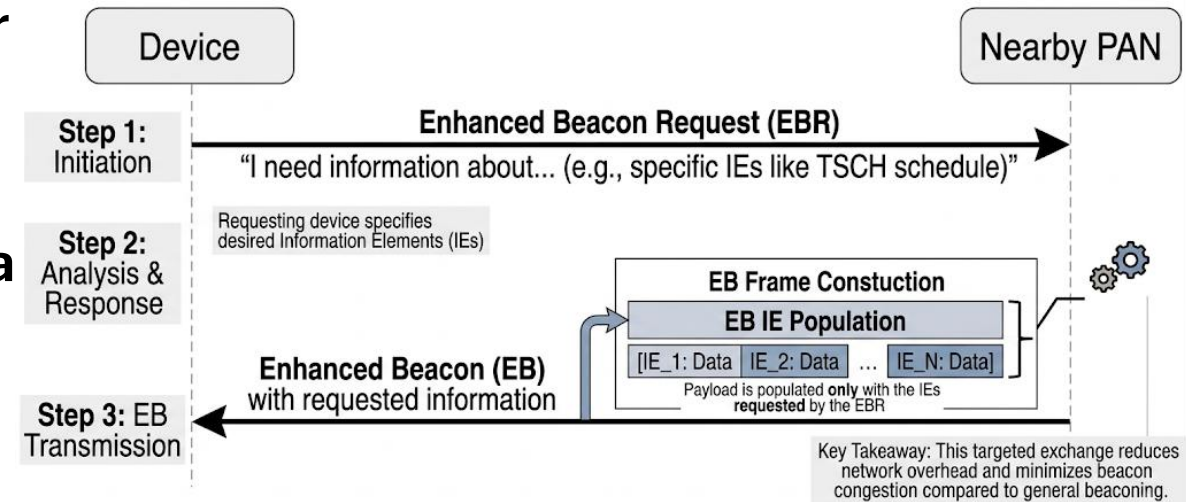
- **Network information** : information about the network/PAN
- **Network metrics** : information about network conditions
- **Frequency-hopping information** : information needed for channel hopping
- **Version information** : identifies the version/configuration of the network

➤ **Why are EBs useful?**

- A normal beacon provides basic network information.
- **Enhanced Beacon = Beacon + Information Elements**
- This makes the beacon **more flexible and extensible**.

IEEE 802.15.4e MAC Enhancements in Details

- **Enhanced Beacon Requests (EBRs)**
- An **Enhanced Beacon Request (EBR)** is a message sent by a device to **ask the network for specific information**.
- Instead of receiving every available piece of network information, the device can **specify what it is looking for**.
- For example, a device may look for a PAN that:
 - **Allows new devices to join**
 - Supports a particular **MAC capability**
 - Supports a particular **PHY capability**
 - Provides specific network information
- This makes **network discovery more selective and efficient**



IEEE 802.15.4e MAC Enhancements in Details

➤ Enhanced Acknowledgment

➤ In wireless communication, when Device A sends a data frame to Device B, Device B sends back a small **Acknowledgement (ACK) frame** to confirm.

➤ Problem with Traditional ACK Frames:

Issue	Explanation
No Frame Counter	Traditional ACK frames do not contain any identifier linking them to the specific data frame being acknowledged
Spoofing Vulnerability	An attacker can send a fake ACK frame to Device A
Consequence	Device A thinks transmission was successful and does not retransmit – resulting in data loss

IEEE 802.15.4e MAC Enhancements in Details

➤ Enhanced Acknowledgment...

- The Enhanced Acknowledgement frame in IEEE 802.15.4e **integrates a frame counter** for the frame being acknowledged.

Feature	Description
Frame Counter	A unique number that identifies the specific data frame being acknowledged
How It Works	When Device B sends an Enhanced ACK, it includes the frame counter of the data frame it just received
Verification	Device A checks that the frame counter in the ACK matches the frame it just sent
Security Benefit	An attacker cannot spoof the ACK because they do not know the correct frame counter value

Security of IEEE 802.15.4e and IEEE 802.15.4g

- Both **IEEE 802.15.4e** and **802.15.4g** use the **security mechanisms of IEEE 802.15.4**.
- Security is mainly provided at the **MAC layer** using **AES-128**.
- Main Security Services

Security service	What it provides
 Confidentiality	Prevents unauthorized users from reading the transmitted data
 Integrity	Detects whether a frame has been modified
 Authentication	Helps verify that a frame comes from a trusted device
 Access control	Prevents unauthorized devices from accessing the network

Security of IEEE 802.15.4e and IEEE 802.15.4g...

Important Security Features

- **AES-128 based cryptography**
- **Frame encryption** for confidentiality
- **Message Integrity Code (MIC)** for integrity/authentication
- **Security keys** are used to protect communication
- **Frame counters** help prevent **replay attacks**
- Security can be enabled selectively for different frames
- 802.15.4 provides a security foundation
- 802.15.4e provides MAC enhancements + security considerations
- 802.15.4g provides PHY enhancements + security inherited from 802.15.4

IEEE 802.15.4g & 802.15.4e – Conclusions

- **Amendments to IEEE 802.15.4** — integrated into **IEEE 802.15.4-2015**
- **802.15.4g: PHY layer**
 - Improves **range, data rate, robustness and frequency flexibility**
 - Designed for **Smart Utility Networks (SUN)**
- **802.15.4e : MAC layer**
 - Improves **reliability, latency and channel access**
 - Includes features such as **TSCH, IEs, EBs and EBRs**
- Together, they address limitations of traditional 802.15.4, including:
 - **High latency**
 - **Multipath fading**
 - Large-scale deployment challenges
- Well suited for:
 -  **Smart Grid / Utilities**
 -  **Smart Cities**
 -  **Large-scale IoT deployments**
- **Wi-SUN Alliance**
 - Promotes **interoperability and certification**
 - Uses **802.15.4g** as a foundation
 - Provides profiles such as **FAN (Field Area Network)**

IEEE 1901.2a

- IEEE 1901.2a is a **wired technology** that updates the original IEEE 1901.2 specification.
- It is the standard for **Narrowband Power Line Communication (NB-PLC)** – a low-frequency PLC solution operating below **500 kHz**
- Designed mainly for **smart grid and utility IoT applications**
- **What is Narrowband Power Line Communication (NB-PLC)?**

Aspect	Description
Medium	Uses existing AC and DC electric power lines for data transmission
Frequency	Operates below 500 kHz
Data Rate	Scalable up to 500 kbps
Range	Supports indoor and outdoor communications over low-voltage (LV < 1000V) and medium-voltage (MV up to 72 kV) power lines

IEEE 1901.2a enables data communication **over existing electrical wiring** – eliminating the need for new cabling in smart grid deployments!

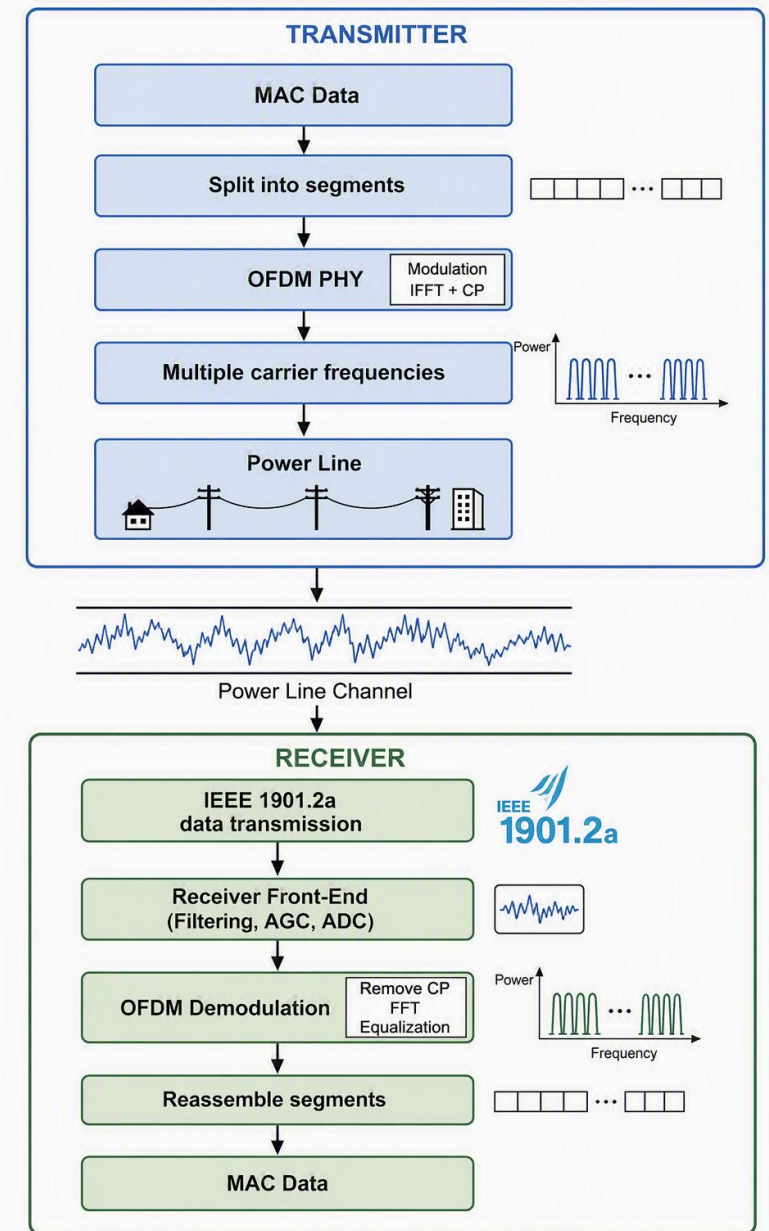
IEEE 1901.2a Standardization and Alliances

Aspect	Details
Standards Body	IEEE
Base Standard	IEEE 1901.2-2013
Amendment	IEEE 1901.2a-2015 – incorporates corrections and clarifications
Industry Alliance	HomePlug Alliance (HomePlug Netricity initiative) – promotes marketing and certification of PLC technology
Key Feature	A worldwide standard for NB-PLC via AC, DC, and non-energized electric power lines

IEEE 1901.2a – PHY Layer

- **Communication medium:** Existing **power lines**
- **Frequency:** 3–500 kHz (Narrowband PLC)
- **Modulation: OFDM**
 - Data is transmitted using **multiple carrier frequencies (tones)** simultaneously
- **Modulation options:**
 - **ROBO** → highly robust, suitable for noisy/poor channels
 - **DBPSK / DQPSK / D8PSK** → provide different data-rate/robustness trade-offs
 - **16-QAM** → higher data rate, available in some bands
- **Data rate: Adaptive**
 - Changes according to **channel conditions and selected modulation**
- **Segmentation:**
 - Large MAC data is divided into **smaller segments** before transmission

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IEEE 1901.2a – MAC Layer

Feature	Description
Frame Format	Based on IEEE 802.15.4 MAC frame
Key Component	Information Elements (IEs) – brought from 802.15.4e
Channel Access	CSMA/CA – Carrier Sense Multiple Access with Collision Avoidance
Additional Capabilities	IEEE 802.15.9 Key Management Protocol, SSID support
Segment Control	Handles segmentation/fragmentation of upper-layer packets larger than MPDU

IEEE 1901.2a – Topology

Topology

- IEEE 1901.2a uses the **existing electrical power-line network** for communication.
- **Supported Network Structures**
- **Star Topology**
 - Devices communicate through a **central coordinator/gateway**.
 - Suitable when devices can directly reach the coordinator.
- **Mesh Topology**
 - Devices can **forward data for other devices**.
 - Provides alternative communication paths.
 - Useful when direct communication with the gateway is difficult.
- **Communication Through the Power Grid**
 - Uses the **existing power distribution network**.
 - Communication can travel across sections of the electrical network.
 - Can support communication **across transformers**, enabling wider-area utility networks.

IEEE 1901.2a – Security

➤ **AES-based Security**

- Uses **AES cryptography** for protecting communication
- Provides **confidentiality, integrity and authentication**

➤ **Protection of Header & Payload**

- Security mechanisms specify **which information in a frame is protected**
- Includes appropriate handling of **Information Elements (IEs)**

➤ **Frame Counter**

- Uses a **frame counter** to help detect and prevent **replay attacks**
- Updated to be consistent with **IEEE 802.15.4e-2012**

➤ **Privacy**

- Protects transmitted information from **unauthorized access**
- Important for sensitive utility data such as **smart-meter readings**

➤ **Security-Sensitive Applications**

- Provides security mechanisms suitable for **critical utility and Smart Grid communication**

IEEE 1901.2a – Security Use Cases

Domain	Application
 Smart Metering	Automated reading and monitoring of electricity, gas and water meters
 Distribution Automation	Monitoring and control of transformers, switches and other grid equipment
 Smart Grid Meter Communication	Communication between utility infrastructure and smart meters
 EV Charging	Communication and monitoring of EV charging infrastructure
 Public Lighting	Remote monitoring, control and scheduling of streetlights
 Microgrids	Monitoring and management of local energy generation, storage and loads
 Renewable Energy	Monitoring of solar, wind and other distributed energy resources
 Smart City Infrastructure	Communication for distributed utility and city infrastructure

IEEE 1901.2a Conclusion

- **Open NB-PLC Standard** – Provides PHY and MAC for communication over existing power lines.
- **Based on Established Technologies** – Leverages **G3-PLC** and **PRIME**.
- **Flexible PHY** – Supports a wide range of **3–500 kHz** frequencies.
- **Feature-Rich MAC** – Based on **IEEE 802.15.4** and supports **mesh networking**.
- **Industry Support** – Supported through **Netricity** and collaboration with **Wi-SUN Alliance**.
- **Key Challenge** – Faces competition from the more established **G3-PLC** and **PRIME** standards.
- **Key Advantage** – Attractive for **end-to-end IP communication** and integration with **IEEE 802.15.4g/e**.

IEEE 802.11ah - Wi-Fi HaLow

- **Amendment to the IEEE 802.11 Wi-Fi standard**
- Operates in **sub-1 GHz** (frequencies below 1 GHz) **license-exempt bands**
- Designed for **long-range, low-power wireless communication**
- Supports **large numbers of IoT devices**
- Marketed as **Wi-Fi HaLow** by the **Wi-Fi Alliance**
- Target applications include **IoT, smart homes, smart cities, and industrial sensing**

Challenge with Traditional Wi-Fi	How 802.11ah Solves It
2.4 GHz and 5 GHz bands are saturated and have limited range	Sub-1 GHz provides better propagation, longer range, and better penetration through obstacles
High power consumption – not suitable for battery-powered IoT	Designed for low power consumption
Limited scalability – few devices per AP	Supports thousands of devices per AP

An access point (AP) is a networking device that bridges wireless devices to a wired local area network

IEEE 802.11ah – Physical Layer (PHY)

- The 802.11ah PHY is designed based on the **down-clocked operation of IEEE 802.11ac's PHY layer**.
- It uses **Orthogonal Frequency Division Multiplexing (OFDM)** modulation

Bandwidth	Purpose
1 MHz	Extra narrow channel – extends range; uses special MCS10 for reliability
2 MHz	Low data rate, low power – ideal for Wireless Sensor Networks (WSN)
4, 8, 16 MHz	Higher data rates – comparable to traditional Wi-Fi

Channel availability varies by region due to licensing limitations

- Data Rates:
 - Minimum Data Rate: ~150 kbps
 - Maximum Data Rate: Up to 347 Mbps
- **Coverage Range:**
 - Can provide coverage **up to 1 km**
 - Sub-1 GHz frequencies penetrate **walls, foliage, and obstacles** better than 2.4/5 GHz

The 1 MHz channel with MCS10 is specifically designed for long-range, low-power IoT applications – making 802.11ah suitable for sensors and smart meters!

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IEEE 802.11ah – MAC Layer Features

- The 802.11ah MAC layer is designed to support **scalability** while keeping **power consumption low**.
- It introduces several new features specifically for IoT
- Key MAC Enhancements:

MAC Feature	Simple Description	Main Benefit
Restricted Access Window (RAW)	AP divides clients into groups and gives each group a specific time to communicate	Reduces collisions – up to 64 time slots
Target Wake Time (TWT)	Devices agree with the AP on when they should wake up and communicate	Saves power – devices sleep at other times
Traffic Indication Map (TIM)	AP tells devices whether they have any data waiting	Saves power – devices do not need to stay awake all the time
Short MAC Header	MAC header is reduced from 28 bytes to 18 bytes	Reduces overhead and improves efficiency
Fast Association/Authentication	Simplifies and speeds up the process of joining the network	Faster network entry for many devices

IEEE 802.11ah – Topology

Aspect	Simple Description
Network Topology	Star topology – all devices (STAs) connect to one Access Point (AP)
Communication	Devices communicate through the AP
Mesh Support	No Layer 2 mesh support
Roaming	Devices can move between APs using other 802.11 features such as 802.11r

IEEE 802.11ah – Security

Aspect	Simple Description
Security Protocols	Supports WPA2 and WPA3 security
Encryption	Uses AES-CCMP (128-bit) to protect wireless data
Authentication	Supports PSK, SAE, OWE, and 802.1X for secure device access
Compatibility	Works with existing Wi-Fi security mechanisms

IEEE 802.11ah Use Cases

Domain	Applications
Smart Cities	Street lighting, waste management, parking sensors
Smart Agriculture	Soil moisture monitoring, irrigation control, livestock tracking
Industrial IoT	Asset tracking, predictive maintenance, environmental monitoring
Utilities	Smart metering (water, gas, electricity), leak detection
Supply Chain	Logistics tracking, cold chain monitoring

LoRaWAN Introduction

- LoRaWAN (Long Range Wide Area Network) is a leading Low-Power Wide-Area Network (LPWAN) protocol designed for scalable IoT applications.
- It is maintained by the LoRa Alliance
- Standardization and Alliances:

Aspect	Details
Standards Body	LoRa Alliance – open, non-profit association
Physical Layer	LoRa – proprietary physical layer from Semtech
MAC Layer	LoRaWAN – open standard maintained by LoRa Alliance
Industry Support	Broad ecosystem of device, gateway, and network server vendors

LoRaWAN Introduction...

Layer	Technology	Description
Physical Layer	LoRa (Semtech proprietary)	Chirp Spread Spectrum (CSS) modulation – provides long-range, robust communication
MAC Layer	LoRaWAN (Open standard)	Defines how devices communicate, device classes, security, and message scheduling

LoRaWAN is an open MAC layer protocol that runs on top of Semtech's proprietary LoRa PHY – enabling long-range, low-power communication for battery-operated sensors

LoRaWAN – Physical Layer (LoRa)

- LoRa uses **Chirp Spread Spectrum (CSS)** modulation – a proprietary technique from Semtech that provides excellent robustness against interference and multipath fading
- **Key PHY Parameters:**

Parameter	Description
Modulation	Chirp Spread Spectrum (CSS) – spreads the signal over a wider frequency range, making it more resistant to noise and interference.
Frequency Band	Uses unlicensed ISM bands: 868 MHz (Europe), 915 MHz (Americas), and 433 MHz (some Asian regions).
Spreading Factor (SF)	SF7–SF12. Higher SF → longer range & better sensitivity, but slower data transmission.
Bandwidth (BW)	125, 250, or 500 kHz. Higher bandwidth → higher data rate, but generally shorter communication range.
Coding Rate (CR)	Adds error-correction bits to the data so the receiver can recover some data even when the signal is weak or noisy.

LoRa: Frequency, Bandwidth & Spreading Factor

Term	What it means	Typical LoRa values
Frequency	Where the radio signal operates	433 / 868 / 915 MHz
Bandwidth (BW)	Width of the frequency range occupied by the signal	125 / 250 / 500 kHz
Spreading Factor (SF)	How much each symbol is spread in time	SF7–SF12

LoRaWAN – MAC Layer Features

- LoRaWAN defines the MAC protocol – how devices communicate with gateways, schedule transmissions, and manage power consumption
- **Key MAC Features:**

Feature	Description
Adaptive Data Rate (ADR)	Network dynamically adjusts SF and transmit power based on signal conditions – optimizes power consumption and spectrum efficiency
Channel Access	Pure ALOHA – devices transmit when they have data (asynchronous)
Bi-directional Communication	Supports both uplink (device to server) and downlink (server to device)
Duty Cycle	Regulated by regional frequency band limitations (e.g., 1% duty cycle in EU)

- Adaptive Data Rate (ADR) is LoRaWAN's most powerful MAC feature
- Network automatically optimizes each device's transmission parameters for minimum power consumption while ensuring reliable message delivery

LoRaWAN – Topology

Aspect	Description
Network Structure	Star-of-Stars topology – end-devices communicate directly with gateways
Gateway Role	Receives LoRa transmissions and forwards them to Network Server (cloud-based)
Network Server	Manages devices, ADR, security, and routes messages to application servers
Scalability	A single gateway can support thousands of end-devices

LoRaWAN – Security

Aspect	Details
Encryption	AES-128 encryption – industry-standard
Authentication	Network Security (MAC layer) – authenticates endpoints with network server
Application Security	Application Security – end-to-end encryption between device and application server
Key Management	Device-specific keys provisioned during onboarding
Security Layers	Two independent security layers: Network (MAC) + Application

LoRaWAN – Competitive Technologies and Use Cases

Technology	Comparison with LoRaWAN
NB-IoT	Cellular LPWAN – licensed spectrum, higher cost, better QoS, similar range
LTE-M	Cellular LPWAN – licensed spectrum, higher data rate, mobility support
IEEE 802.11ah (Wi-Fi HaLow)	Higher throughput, shorter range, more power consumption
SigFox	Ultra-narrowband – even lower data rate, simpler, more limited downlink

LoRaWAN – Use Cases

- **Smart Cities:** Street lights, smart parking, waste management
- **Smart Agriculture:** Soil monitoring, irrigation, livestock tracking
- **Industrial IoT:** Asset tracking, equipment monitoring, predictive maintenance
- **Utilities:** Smart water/gas/electricity meters, leak detection
- **Supply Chain:** Shipment tracking, cold-chain monitoring

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The Need for Optimizing IP in IoT

➤ IoT Devices are Resource-Constrained

- Limited **memory, processing power, and energy**
- **Class 0:** Highly constrained → needs IP adaptation
- **Class 1:** Moderately constrained → needs optimized IP
- **Class 2:** More capable → can use full IP stack

➤ IoT Networks are Constrained

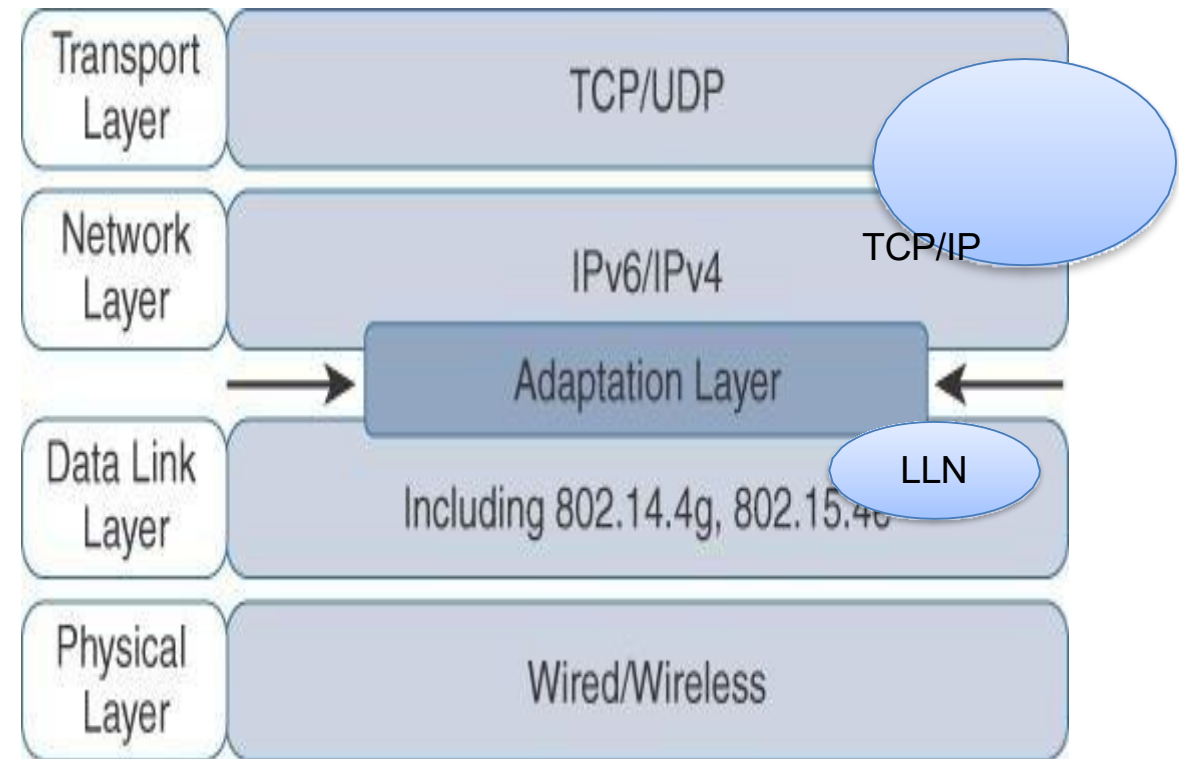
- **Low bandwidth** and high packet loss
- Unreliable communication links
- **Small frame size** (IEEE 802.15.4: 127 bytes)

➤ IPv6 Header Overhead

- IPv6 header = **40 bytes**
- IEEE 802.15.4 frame = **127 bytes**
- Small IoT data → **large overhead**
- Example: **10-byte data + 40-byte IPv6 header**

6LoWPAN

- IETF formed 6LoWPAN WG in 2004 to design the Adaptation Layer
- **6LoWPAN**: IPv6 over Low-power Wireless Personal Area Networks
- How to carry IPv6 packet efficiently within small link layer frames such as IEEE 802.15.4?

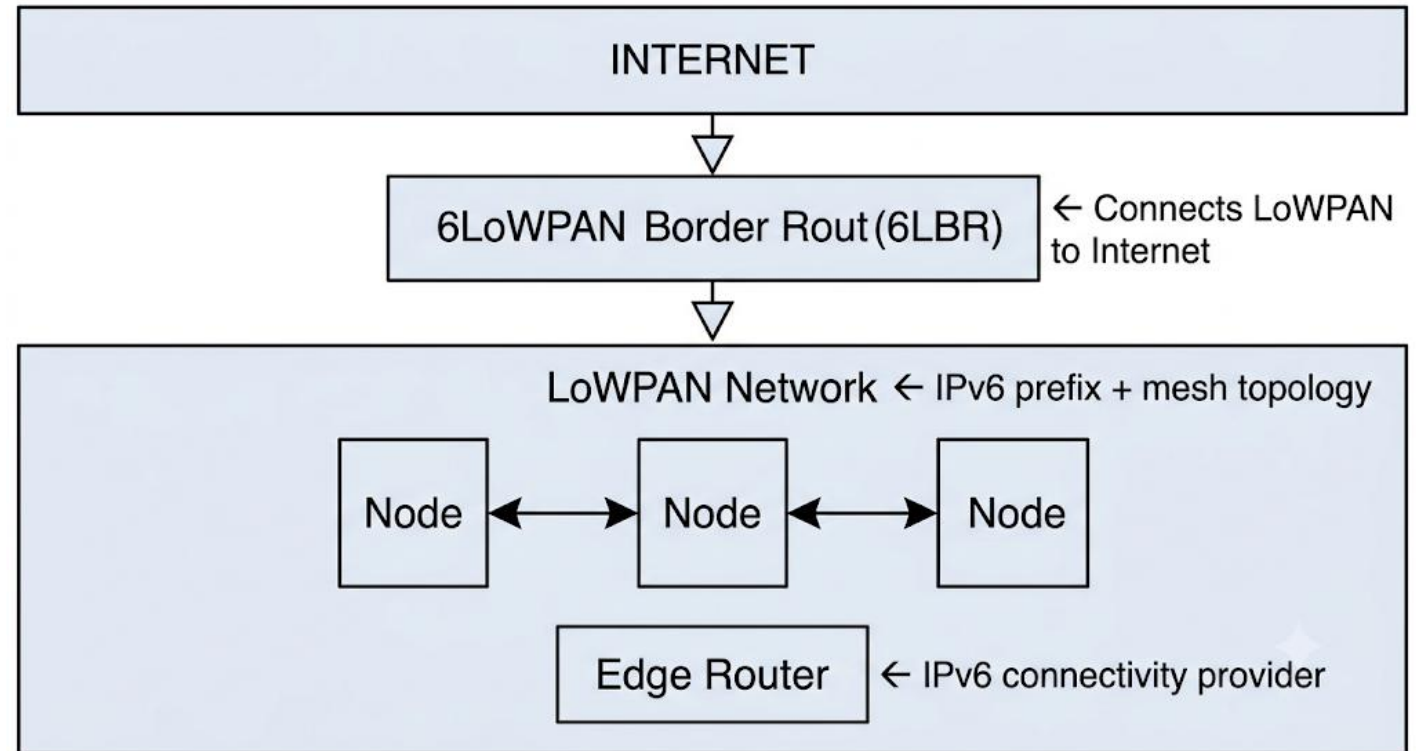


- IPv6 MTU to be **at least 1280 bytes** in length.
- 802.15.4's standard packet size of **127 octets**.

- **Goal**: Enable IP/IPv6 communication even for highly constrained IoT devices.
- **Target Devices**: Low-power devices with limited memory, processing power, and energy.
- **Key Functions**:
 - **IPv6 Header Compression** → reduces communication overhead
 - **Packet Fragmentation & Reassembly** → supports small frame sizes
 - **Adaptation Layer** → connects IPv6 with low-power networks such as IEEE 802.15.4
- **Applications**: Home Automation, Smart Grid, Industrial Monitoring, Smart Agriculture, Building Automation

6LoWPAN Architecture

- **6LoWPAN Architecture**
- **Internet:** Connects the LoWPAN to IPv6 networks, servers, and cloud.
- **6LoWPAN Border Router (6LBR):** Connects the LoWPAN ↔ Internet and handles IPv6 adaptation.
- **LoWPAN Network:** Low-power IPv6 network using technologies such as IEEE 802.15.4.
- **Nodes:** Sensors and actuators that communicate over a low-power mesh network.
- **Edge Router:** Provides routing and IPv6 connectivity within the local network.



What is Low-Power and Lossy Network?

- **Constrained Node:** A node where some of the characteristics that are otherwise pretty much taken for granted for Internet **nodes** are not attainable, often due to cost constraints and/or physical constraints on characteristics such as size, weight, and available power and energy.

- tight limits on **power**, **memory**, and **processing resources**

Low-Power and Lossy Network (LLN)

- A network of **resource-constrained devices** with limited:
 - **Power/energy**
 - **Memory**
 - **Processing capability**
- Uses **low-power wireless links**, such as **IEEE 802.15.4** and low-power Wi-Fi.
- Communication links are often **unreliable and lossy**, causing:
 - **Low data rates**
 - **Packet loss**
 - **Variable connectivity**
- Commonly used in **IoT and sensor networks**.

Routing challenges in LLNs

LLN Characteristic	Challenge for Routing
Limited computational power	Cannot run complex routing algorithms
Limited storage/memory	Cannot maintain large routing tables
Low transmission range	Requires multi-hop communication
Battery-powered	Must minimize energy consumption
High loss rates	Unreliable links cause frequent retransmissions
Low data rates	Overhead must be minimized

Can't use OSPF, OLSR, RIP, AODV, DSDV, DSR, etc

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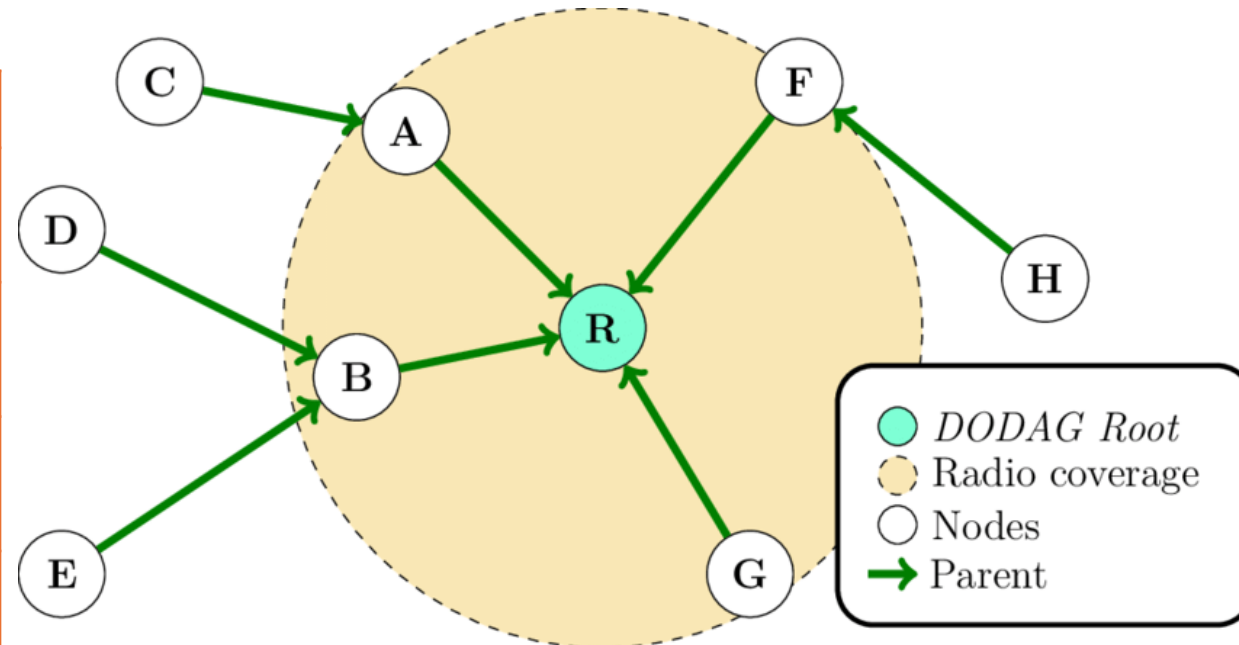
RPL – Introduction

- RPL (IPv6 Routing Protocol for **LLN**) is a distance-vector routing protocol designed by the IETF for resource-constrained IoT networks.
- Standardized in RFC 6550.
- **Why LLNs Need a Specialized Protocol**
- Low-Power and Lossy Networks (LLNs) have unique characteristics that make traditional routing protocols unsuitable

RPL – Core Concepts and DODAG

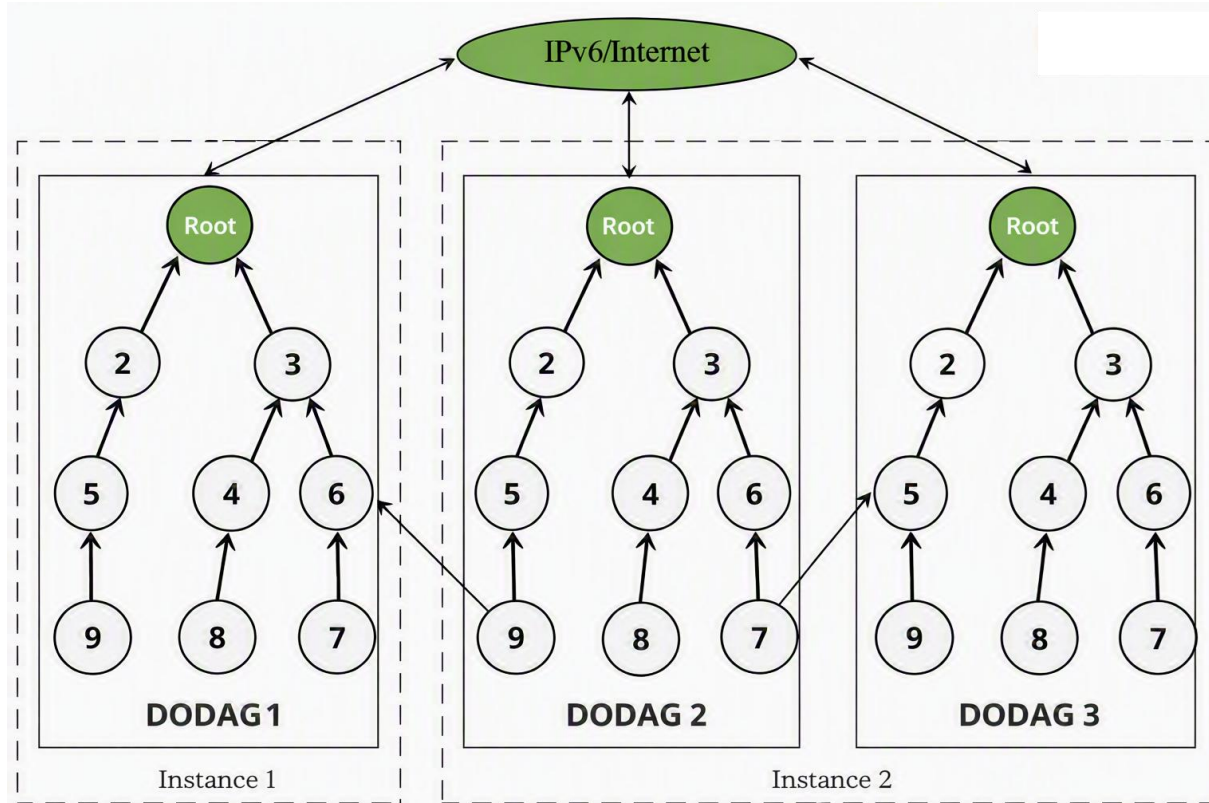
- RPL organizes the network as a **Destination-Oriented Directed Acyclic Graph (DODAG)** – a tree-like topology with no cycles, oriented toward a single destination (root node).

Term	Description
DODAG	Destination-Oriented Directed Acyclic Graph – the complete network topology
Root	The single destination node (sink/gateway) that all traffic flows toward
Rank	A numerical value indicating a node's distance from the root
Preferred Parent	The best next-hop node selected by each router to reach the root



RPL Instance

- An RPL Instance is a set of one or more DODAGs that **share a common RPLInstanceID**



RPL Instances

- **RPLInstanceID** is a **unique** identifier within a network.
- DODAGs with the same RPLInstanceID share the same Objective Function (OF)
 - used to compute the position of node in the DODAG .
- An **objective function** (OF) defines
 - how metrics are used to select **routes** and establish a node's **rank**.
- Objective Function computes the “**rank**”
 - rank is the “distance” between the node and
 - DODAG root
 - *Rank* should monotonically decrease along the DODAG and towards the destination

IoT Application Transport Methods

- IoT application transport methods are the communication protocols and mechanisms used to move data between smart devices and cloud or local applications

Transport Method Categories

➤ Legacy Industrial Protocols (Adapted for IP)

- **SCADA, DNP3, Modbus, IEC 60870-5-104** – originally serial-based, now adapted to run **over IP** networks via TCP/UDP
- **DNP3** (IEEE 1815) and **Modbus/TCP** are commonly used in utilities, building management, and transportation

➤ Generic Web Protocols

- **HTTP/HTTPS** – used on consumer/enterprise IoT devices with sufficient resources; HTTPS adds TLS security
- **TCP/UDP** – foundational transport layers; TCP reliable, UDP connectionless and low-overhead

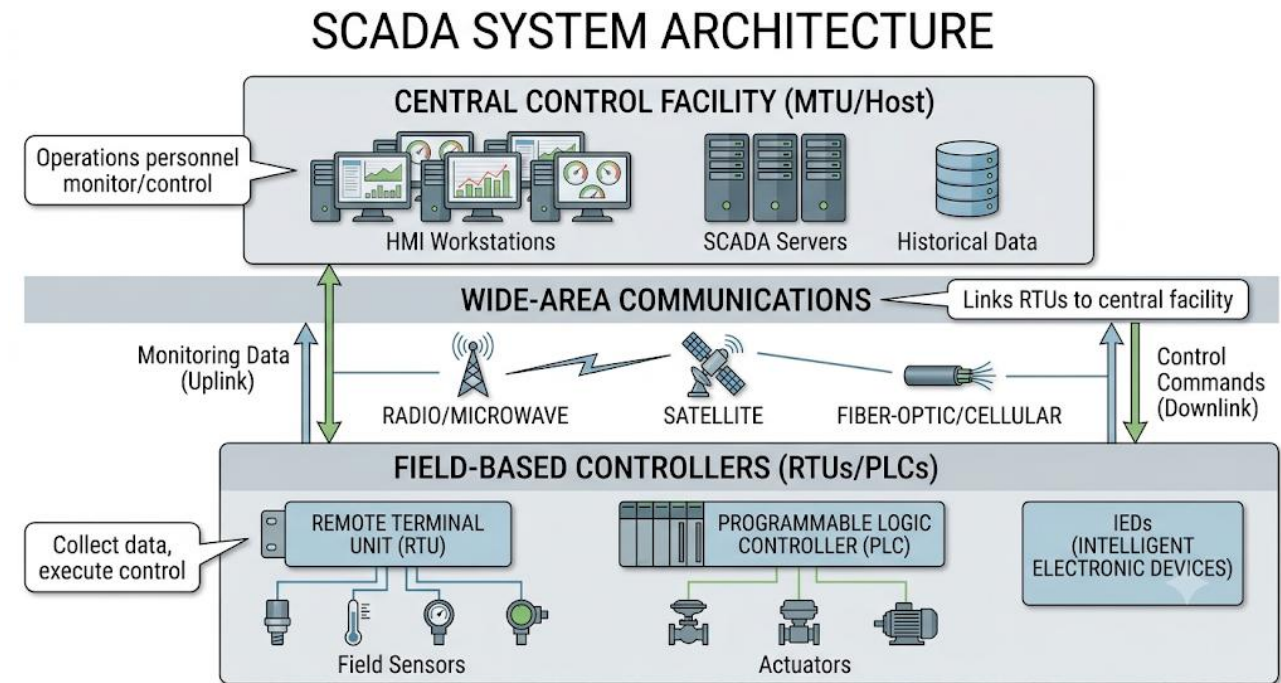
➤ Lightweight IoT Protocols (Optimized for Constrained Nodes)

- **MQTT** – lightweight publish/subscribe over TCP; supports QoS levels for reliability; widely used for telemetry
- **CoAP** – RESTful request/response over UDP; very low overhead (4-byte header); designed for constrained networks; supports observation
- **AMQP** – feature-rich enterprise messaging (queuing, routing, reliability)
- **DDS** – real-time publish/subscribe for high-performance industrial applications
- **LwM2M** – device management + telemetry using CoAP transport

Supervisory Control and Data Acquisition (SCADA)

- Supervisory Control and Data Acquisition (SCADA) is a centralised system that enables remote monitoring, control, and data acquisition from field devices across large geographic areas .
- It is one of the key application transport methods for industrial IoT and critical infrastructure

Component	Description
RTU (Remote Terminal Unit)	Field-based controller that interfaces with sensors/actuators
MTU (Master Terminal Unit) / Host Computer	Central control facility for operators
Communications Network	Wide-area links connecting RTUs to central facility



Common SCADA Communication Protocols

Protocol	Description
Modbus/TCP	Widely used serial/IP protocol for industrial devices
DNP3 (Distributed Network Protocol)	Standard for utility/SCADA communication
IEC 60870-5-101/104	Telecontrol protocols for power systems
IEC 61850 MMS	For substation automation and reporting
TCP/IP	Enables IP-based RTU connectivity

- Modern RTUs are increasingly IP-enabled, supporting protocols like Modbus/TCP and DNP/IP over TCP/IP networks

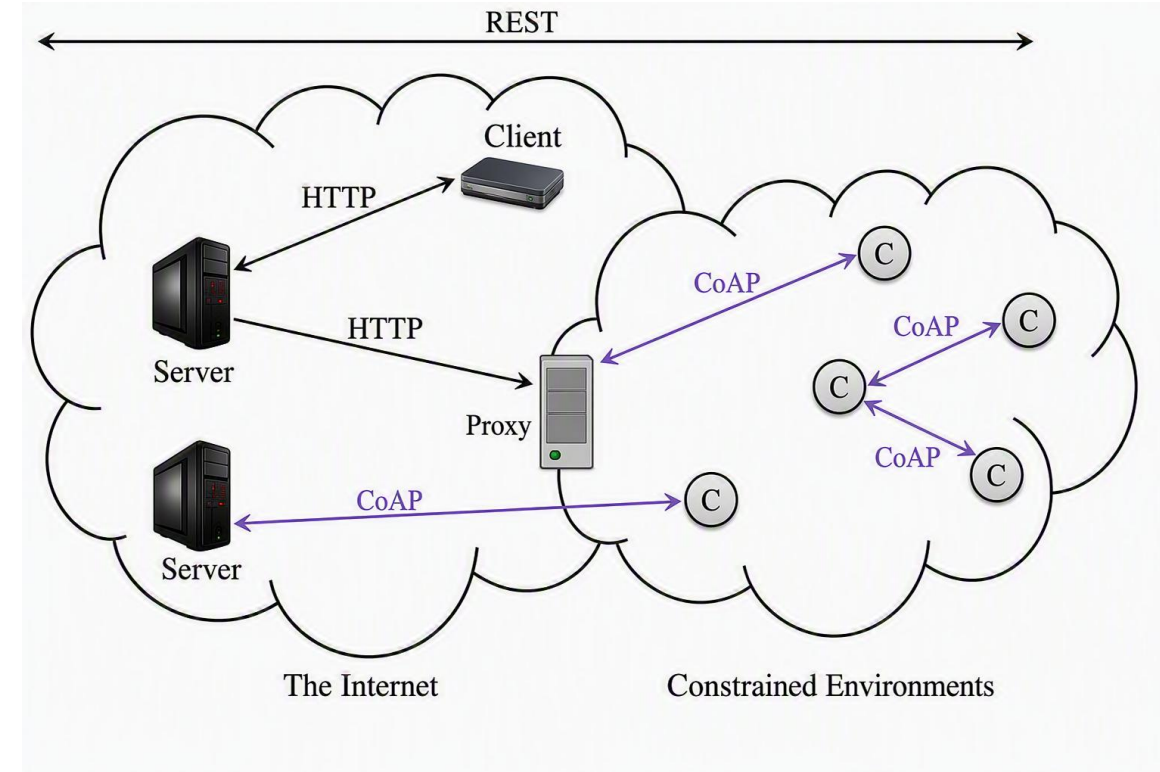
IoT Application Layer Protocols

- IoT Application Layer Protocols are specialized communication protocols that enable data exchange between IoT devices and applications in constrained environments.
- They sit at the top of the IoT protocol stack, governing how applications interpret and act upon sensor data
- **Two Dominant Protocols**

Protocol	Full Form	Standard
CoAP	Constrained Application Protocol	IETF RFC 7252 (2014)
MQTT	Message Queuing Telemetry Transport	OASIS Standar

CoAP Introduction

- CoAP - Constrained Application Protocol.
- **Specialised** web transfer protocol.
- Devised for **constrained** and **low power** networks.
- CoAP is an **application layer protocol** (similar to HTTP).
- Inspired by HTTP, it follows the **request-response** pattern
- CoAP has a **transparent mapping** to HTTP.
- The CoAP protocol spec is specified in **RFC 7252**.
- It is a very efficient **RESTful** protocol.



CoAP Properties

- CoAP is a **specialized web transfer protocol** for use with constrained nodes and constrained (e.g., low-power, lossy) networks.
- It is designed for M2M applications such as smart energy and building automation
- **Key Features**

Feature	Description
RESTful Model	Provides request/response interaction model between application endpoints
HTTP Integration	Designed to easily interface with HTTP for integration with the Web
Built-in Discovery	Supports built-in discovery of services and resources
Multicast Support	Native multicast support for group communication
Very Low Overhead	Minimal 4-byte header
UDP Transport	Runs over UDP by default (also TCP, WebSockets)

CoAP – Protocol Structure

- CoAP consists of **two different layers** that work together to enable lightweight communication in constrained IoT environments:
- **Messages Layer** – Handles UDP-based message exchanges, reliability, and retransmission
- **Request/Response Layer** – Manages application-level request/response interactions
- **Message Types (Messages Layer)**

Message Type

CON (Confirmable):

NON (Non-Confirmable):

ACK (Acknowledgement):

RST (Reset):

Description

Requires an ACK; if no ACK received, message is retransmitted

Does NOT require ACK; no retransmission

Confirms receipt of a CON message

Indicates rejection or error (response to a request)

CoAP – Protocol Structure

➤ Request/Response Layer

Component Description

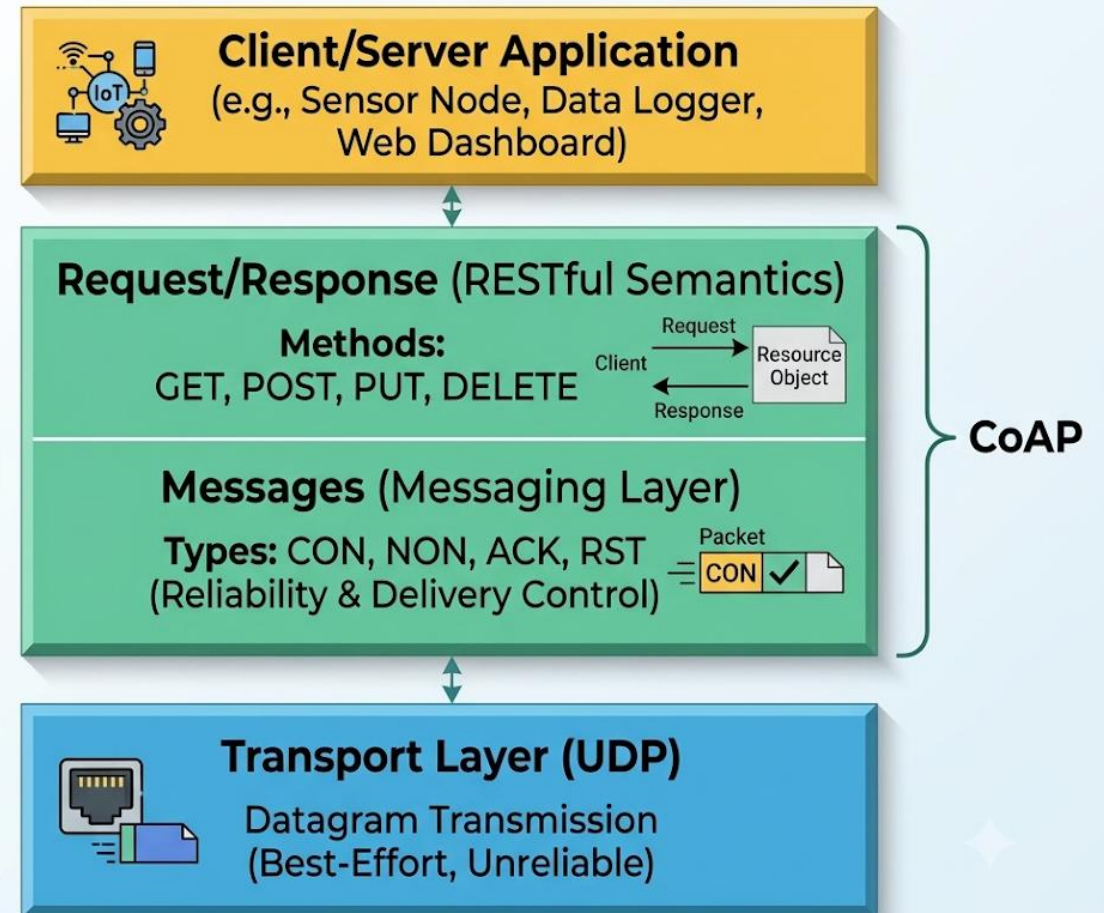
Request Methods:	GET (retrieve), POST (create/update), PUT (replace), DELETE (remove)
Response Codes:	Success (2.xx), Client Error (4.xx), Server Error (5.xx)

➤ How They Work Together

- The **Messages Layer** handles asynchronous UDP communication and re-transmitting lost packets
- The **Request/Response Layer** manages request/response interaction using the underlying message layer

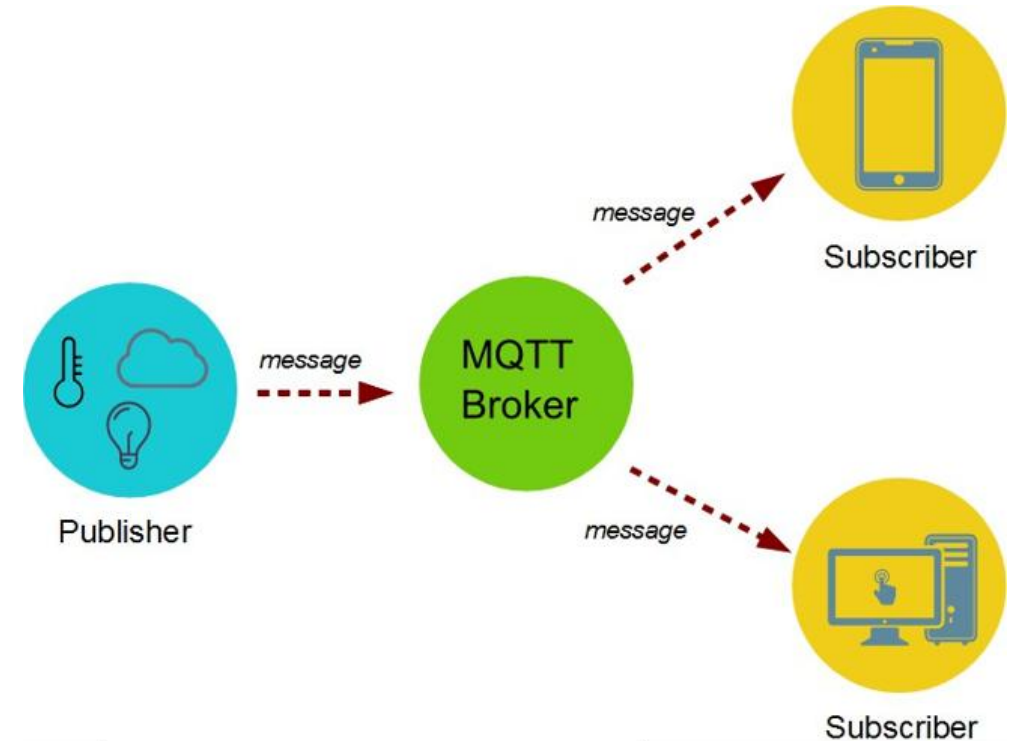
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Constrained Application Protocol (CoAP) Layered Architecture



Message Queuing Telemetry Transport (MQTT)

- MQTT : **M**essage **Q**ueueing **T**elemetry **T**ransport protocol.
 - **Reliable, Lightweight, Cost-effective protocol** that transports messages between devices.
 - Suited for the **transport of telemetry data** (sensor and actuator data)
- Invented by *Andy Stanford Clark* of IBM and *Arlen Nipper* of Arcom (now Eurotech) in 1999
- Used by real-life IoT frameworks:
 - Amazon Web Services (AWS),
 - IBM WebSphere MQ,
 - Microsoft Azure IoT,
 - Facebook Messenger,
- **Example:**
 - **Light sensor** continuously **publish** sensor data to the **broker**.
 - **Building control application** **subscribes** to light sensor data and so receives it from the broker. Then it decides to activate **Camera**.
 - **The application** sends an activation message to the **camera** node (i.e. actuator) through the **broker**



MQTT Features

Feature

Description

Publish/Subscribe Model:

Decouples message publishers from subscribers

Broker Architecture:

Central broker handles message routing and distribution

TCP Transport:

Runs over TCP for reliable, ordered delivery

Very Small Header:

Minimal 2-byte header

Three QoS Levels:

Configurable delivery reliability

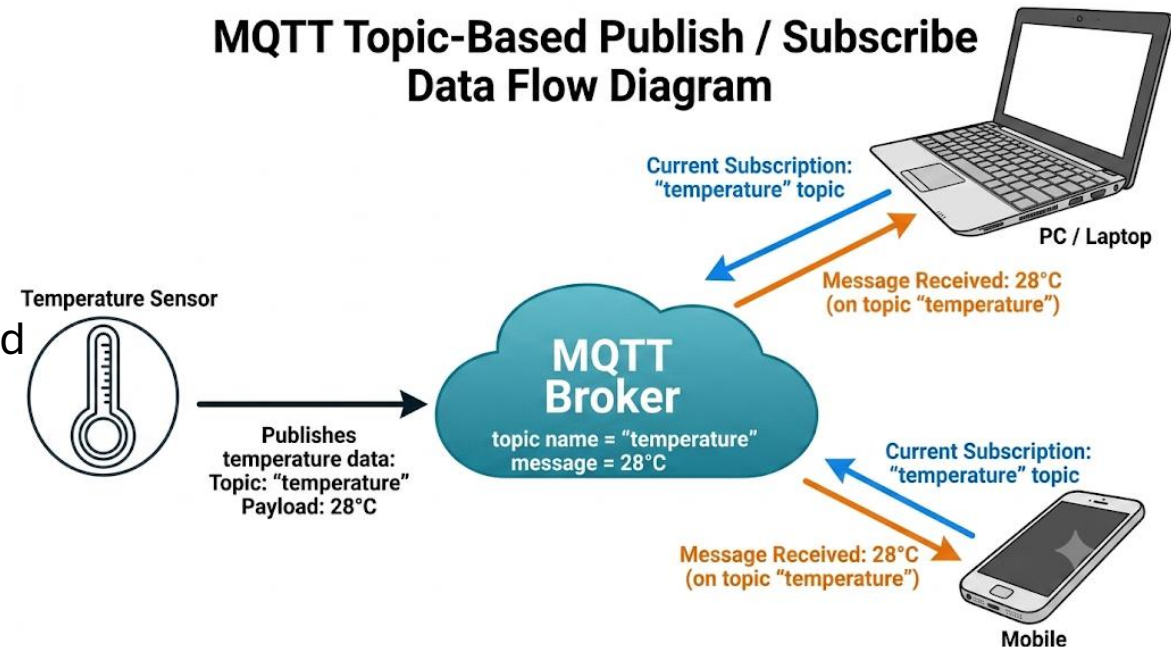
Keep-Alive:

Built-in ping mechanism to maintain connection

- **Publisher** – Device that transmits data as a "topic"
- **Subscriber** – Device that receives topic information
- **Broker** – Facilitates communication; distributes data to all subscriber

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MQTT Topic-Based Publish / Subscribe Data Flow Diagram



MQTT – Quality of Service (QoS) Levels

- MQTT provides three QoS levels for message delivery assurance, allowing developers to choose the appropriate reliability for their application

QoS Level	Name	Delivery Guarantee	How It Works	Use Case
0	At Most Once	Message may be lost	Publisher sends without acknowledgement	Non-critical sensor data (temperature, humidity)
1	At Least Once	Message delivered at least once; possible duplicates	Publisher receives ACK from broker	Critical alerts (security, alarms)
2	Exactly Once	Message delivered exactly once	4-step handshake: publish→receive→release→complete	Financial transactions, billing data

CoAP vs MQTT – Comparison

Aspect	CoAP	MQTT
Transport Layer	UDP (default)	TCP
Header Size	4 bytes	2 bytes
Communication Model	Request/Response (RESTful)	Publish/Subscribe
Security	DTLS	TLS
Certificate Chain	~400 bytes	~4 kB
Authentication	Token-based	Client certificate-based
Message Reliability	Lower (UDP)	Very High (TCP + QoS)
Best For	One-to-one, resource-constrained devices	One-to-many, scalable deployments